

Paul R. Glaum

Applied Mathematical Modeler & Env. Data Scientist

Ann Arbor, MI

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[Google Scholar](#) [Research Gate](#) [GitHub](#)

Summary: PhD ecologist and evolutionary biologist. 10+ years producing novel and high impact research with data-driven results across multiple fields. Expertise in project design, project management, multiple programming environments, applied mathematics, statistical analysis, and interpreting complex scientific results for general audiences. Organized, motivated, & adaptable to evolving project objectives while working both independently and in teams.

Competencies

Core Skill Set	Model building & analysis; Experimental design; Data management, manipulation, analysis, & visualization; Scientific communication,
Types of Models	ODEs, PDEs, Networks, IBMs (agent based models), Discrete time models, Spatial genetic models, Stochastic models, Bayesian Networks, Toxicological models, etc.
Analysis	Numerical analysis, Network analysis, Machine learning (neural networks, random forests, etc.), Frequentist & Bayesian statistics, Bifurcation analysis, Large databases & data warehouses (data cleaning, curation, ETL, etc.), EDA.
Types of Data	Network, Time series, Spatiotemporal, Environmental, Socio-economic, GIS raster data, Species distributions, Allometric/Metabolic, Organismal traits, Allele frequencies, etc.
Programming	R, Matlab, Mathematica, Linux, Command line, JMP, SQL, Python, Java, C++, HTML
Computing	Cloud/cluster/HPC computing, pbs/slurm scripting, Globus, Git (Github), etc.

Professional Experience

Present June 2022	Waterborne Environmental Inc. – Senior Scientist/Ecological Modeler • Client facing risk analyst covering chemical run-off, ecotoxicology, and mitigation. • Co-lead stats & data team. Assisted federal grant writing process. • Engage w/ multiple regulatory models, regulatory agencies, industry groups
May 2022 May 2020	U. of California – Davis, Environmental Science & Policy: Postdoctoral Scholar • Wrote computationally efficient method for integrating 100s of unique organismal phenologies into large network models using the replicator equation. • Created novel tests and network statistics to corroborate model output with a large empirical, field, and museum dataset (curated by collaborators and myself). Article Link • Developed novel extension of machine learning techniques into analysis of ODE simulation data while mentoring graduate student research. Article Link
May 2020 May 2018	University of Michigan, Ecology & Evolutionary Biology/ Michigan Computational Discovery and Engineering: Postdoctoral Res. Fellow • Integrated economic and biological models with fishery yield data to develop first network based bio-economic model of fishery dynamics. Article Link • Analyzed large simulation datasets using machine learning and regression techniques. • Collaborative analysis of historical pollinator community database using museum specimen records, contemporary sampling, and neural networks. Article Link
May 2018 Sept 2012	University of Michigan, Ecology & Evolutionary: Grad Student Researcher • Independently funded & implemented longitudinal pollinator field research integrating land cover and socio-economic GIS data. Mentored & trained 10 undergraduate students. • Contacted by multiple news outlets to discuss findings, e.g. NPR, PBS, Sierra Club, Detroit Metro Times, The Scientist Magazine, Next City • Developed and analyzed multiple ODE models vetted with empirical data. Article Link
Multiple Years	Teaching Positions • GRAD. INSTRUCTOR: U of Mich • ESL Teacher: Shimane Prefecture, Japan.

Education

PhD University of Michigan – Ecology & Evolutionary Biology
MSc. University of Michigan – Ecology & Evolutionary Biology
B.S. University of Wisconsin, Madison – Mathematics & Japanese

Additional: Conversational Japanese, Grant/Proposal Writing, [Conference](#) organizer, Public speaking (20+ presentations), Mentoring